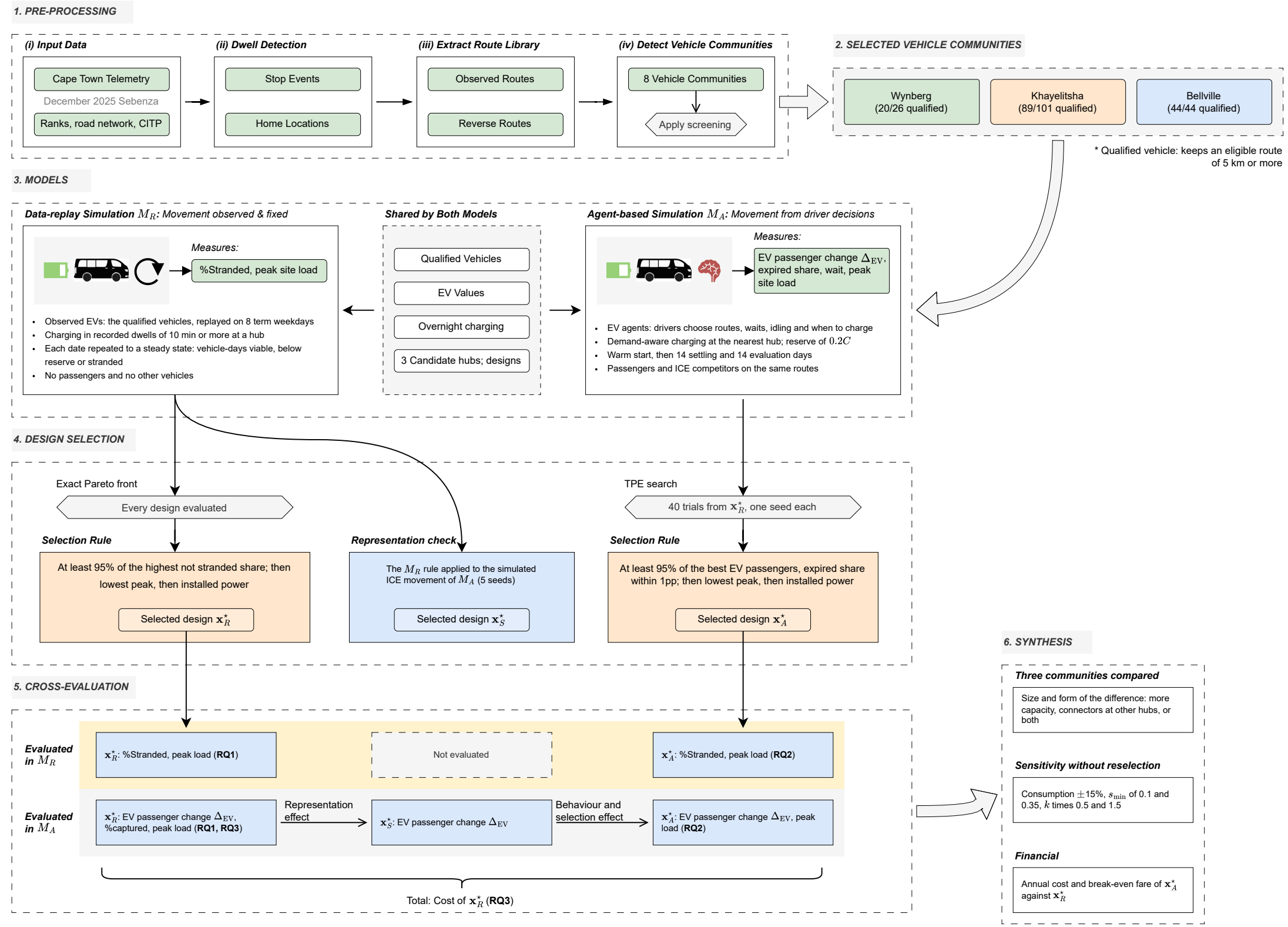

1. PRE-PROCESSING
(i) Input Data
Cape Town Telemetry
December 2025 Sebenza
Ranks, road network, CITP
(ii) Dwell Detection
Stop Events
Home Locations
(iii) Extract Route Library
Observed Routes
Reverse Routes
(iv) Detect Vehicle Communities
8 Vehicle Communities
Apply screening
2. SELECTED VEHICLE COMMUNITIES
Wynberg
(20/26 qualified)
Khayelitsha
(89/101 qualified)
Bellville
(44/44 qualified)
* Qualified vehicle: keeps an eligible route of 5 km or more
3. MODELS
Data-replay Simulation M_R: Movement observed & fixed
Measures:
%Stranded, peak site load
Observed EVs: the qualified vehicles, replayed on 8 term weekdays
Charging in recorded dwells of 10 min or more at a hub
Each date repeated to a steady state: vehicle-days viable, below reserve or stranded
No passengers and no other vehicles
Shared by Both Models
Qualified Vehicles
EV Values
Overnight charging
3 Candidate hubs; designs
Agent-based Simulation M_A: Movement from driver decisions
Measures:
EV passenger change Δ_{EV}, expired share, wait, peak site load
EV agents: drivers choose routes, waits, idling and when to charge
Demand-aware charging at the nearest hub; reserve of $0.2C$
Warm start, then 14 settling and 14 evaluation days
Passengers and ICE competitors on the same routes
4. DESIGN SELECTION
Exact Pareto front
Every design evaluated
Selection Rule
At least 95% of the highest not stranded share; then lowest peak, then installed power
Selected design $\mathbf{x}_R^\star$
Representation check
The M_R rule applied to the simulated ICE movement of M_A (5 seeds)
Selected design $\mathbf{x}_S^\star$
TPE search
40 trials from $\mathbf{x}_R^\star$, one seed each
Selection Rule
At least 95% of the best EV passengers, expired share within 1pp; then lowest peak, then installed power
Selected design $\mathbf{x}_A^\star$
5. CROSS-EVALUATION
Evaluated in M_R
$\mathbf{x}_R^\star$: %Stranded, peak load (RQ1)
Not evaluated
$\mathbf{x}_A^\star$: %Stranded, peak load (RQ2)
Evaluated in M_A
$\mathbf{x}_R^\star$: EV passenger change Δ_{EV}, %captured, peak load (RQ1, RQ3)
Representation effect
$\mathbf{x}_S^\star$: EV passenger change Δ_{EV}
Behaviour and selection effect
$\mathbf{x}_A^\star$: EV passenger change Δ_{EV}, peak load (RQ2)
Total: Cost of $\mathbf{x}_R^\star$ (RQ3)
6. SYNTHESIS
Three communities compared
Size and form of the difference: more capacity, connectors at other hubs, or both
Sensitivity without reselection
Consumption $\pm 15\%$, s_{min} of 0.1 and 0.35, k times 0.5 and 1.5
Financial
Annual cost and break-even fare of $\mathbf{x}_A^\star$ against $\mathbf{x}_R^\star$